

Cheap Multi-Robot Autonomous Fleets for Disaster Recovery: Phone-as-Brain Rovers with Depth Sensing, Obstacle Avoidance, and Fleet Command Interfaces

Yojan Gautam

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Abstract

This thesis investigates low-cost multi-robot autonomous fleets for disaster recovery and related high-stakes settings such as military surveillance and fire and rescue. Each rover treats a consumer mobile phone as the primary sensor, brain, and compute node: an operator downloads an app and mounts the phone on a simple chassis, rather than provisioned industrial compute. Terra is that rover and runs local autonomy. ARGOS is the fleet operator layer: it sends high-level missions and waypoints, and it schedules which vehicle needs human attention. The work covers depth sensing and local obstacle avoidance on the vehicle, multi-robot coordination primitives, and that operator interface so one person can supervise a fleet. The author has implemented the autonomy stack and human interface from the ground up; a rolling chassis with depth sensing and obstacle avoidance is currently in progress. The goal is a practical, affordable path from phone-mounted robots to one-to-many fleet supervision in environments where existing platforms are too expensive, brittle, or operationally heavy.

Contents

1	Motivation	2
2	Related Work	2
3	Platform	2
4	Autonomy	2
5	Human Interface	3
6	Evaluation Across Scenarios	3

1 Motivation

Disaster response environments are cluttered, time-critical, and often unsafe for continuous human presence, yet most capable robots remain expensive, logistics-heavy, or tightly coupled to specialized compute. This section argues why conventional platforms underperform for rapid, large-scale recovery and surveillance, and introduces the phone-as-brain insight: a commodity smartphone already packs cameras, IMU, networking, and enough compute to host navigation and sensing software when mounted on a cheap chassis. It also frames secondary applications in military surveillance and fire and rescue, where the same cost and deployability constraints apply. Closing paragraphs state the research question around the minimum per-robot autonomy needed for one human to supervise many vehicles. The single-rover feasibility case is drafted as Paper 1 in `papers/paper1-phone-as-brain/`.

2 Related Work

This section situates the thesis among local motion planners, autonomy taxonomies for fleets, and immersive command interfaces. It reviews Dynamic Window Approach (DWA) and related local planners for differential-drive obstacle avoidance [1], surveys fleet autonomy levels from waypoint following to coordinated multi-robot behavior [2, 3], and summarizes AR and VR command-center designs for multi-robot supervision [4, 5]. Gaps highlighted include cost barriers, phone-centric compute stacks, and end-to-end systems that jointly treat onboard avoidance and human fleet interfaces.

3 Platform

The platform chapter describes the physical and software stack that makes a phone-mounted rover practical. It covers the rolling chassis, mounting of the phone as the main sensor and compute node, the downloadable app that turns the handset into the vehicle brain, and depth sensing used for near-field geometry. Terra owns sensing, mapping, planning, and drive. AR-GOS owns fleet commands and attention scheduling, and does not stream low-level velocity as the operator contract; go-to-waypoint is the intended command, and a twist topic is only a temporary debug path. Implementation status is stated clearly: autonomy and interface software built ground-up by the author, with a rolling chassis, depth sensor, and obstacle avoidance currently progressing toward a repeatable demo.

4 Autonomy

Autonomy spans local obstacle avoidance through multi-robot coordination. At the vehicle level, the section details depth-based obstacle representation, differential-drive kinematics, and DWA-style local planning that replans every cycle so sequential obstacles update the trajectory without a single precomputed arc [1]. At the fleet level, it describes how increased onboard autonomy shifts the human from continuous teleoperation toward exception handling, including shared goals, conflict avoidance, and coordination messages between robots. Open work

(integration with waypoint navigation, multi-obstacle replanning) is documented as part of the experimental roadmap.

5 Human Interface

The human interface chapter presents the ARGOS dashboard for one operator seat, including attention-needed indicators and map-centric views of each rover’s pose and state. An urgency classifier ranks which vehicle needs the person first. HUD, VR, and AR layouts are interface options around that scheduler, not a replacement for it. The narrative ties interface design to autonomy level: richer onboard avoidance should reduce continuous control load and increase the practical fleet size one person can supervise.

6 Evaluation Across Scenarios

Evaluation plans cover single-robot obstacle courses, sequential multi-obstacle navigation, and multi-robot scenarios relevant to disaster recovery, fire and rescue, and surveillance-style coverage. The primary factors are fleet size N and autonomy level A . Candidate metrics include waypoint success, clearance, interventions per robot, operator response time to an attention request, task completion, and NASA-TLX workload as N and A vary. This section also outlines expected failure modes (depth noise, actuation calibration limits, interface attention bottlenecks) and what would constitute a convincing demo versus a controlled study.

References

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